

- 4 (a) (i) At the top of the circle,

$$T_T + mg = \frac{mv_T^2}{L}$$

For the ball to just complete the vertical circle, the tension T_T at the top of the circle is zero.

$$mg = \frac{mv_T^2}{L}$$

$$v_T^2 = gL$$

$$v_T = \sqrt{gL}$$

- (ii) By the principle of conservation of energy, as the ball moves from the top to the bottom of the circle, its gravitational potential energy decreases and its kinetic energy increases. This means the speed at the bottom is greater than the speed at the top. Hence $\frac{v_B}{v_T} > 1$.

- (iii) $\frac{v_B}{v_T} = 3$
 $v_B = 3v_T$

As the ball moves from the top to the bottom,
 increase in K.E. = decrease in G.P.E.

$$\frac{1}{2}mv_B^2 - \frac{1}{2}mv_T^2 = mg(2L)$$

$$\frac{1}{2}m(3v_T)^2 - \frac{1}{2}mv_T^2 = mg(2L)$$

$$\frac{9}{2}mv_T^2 - \frac{1}{2}mv_T^2 = 2mgL$$

$$4mv_T^2 = 2mgL$$

$$v_T = \sqrt{\frac{1}{2}}\sqrt{gL}$$

As $\sqrt{\frac{1}{2}}\sqrt{gL} < \sqrt{gL}$, where $\sqrt{gL}$ is the value of v_T at which the string just goes slack,

the ball will not be able to complete a full circle if $\frac{v_B}{v_T} = 3$. Hence, $\frac{v_B}{v_T} = 3$ is not possible to achieve.

- (b) Considering forces along the radial direction,

$$T \sin \theta = mr\omega^2$$

Since $r = L \sin \theta$,

$$T \sin \theta = m(L \sin \theta)\omega^2$$

$$T = mL\omega^2$$

Since m and L are constants, $T \propto \omega^2$. Hence when the angular velocity is doubled, the tension in the string is $4T$.

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5. (a) (i) Gravitational potential at a point in a gravitational field is the work done per unit